ReflectAI

AI-Powered Retroreflectivity Measurement System

Automated Vision-Based Assessment of Road Signs, Pavement Markings, Road Studs & Delineators Under All Environmental Conditions

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Solution Type:	AI/ML + Computer Vision
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CUTIEFR

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Contents

1 Executive Summary **2**

2 Problem Statement **2**
2.1 Background: What is Retroreflectivity? . . . 2
2.2 The Scale of the Problem . . . 3
2.3 Limitations of Current Handheld Methods . . . 4

3 Proposed Solution **4**
3.1 Core Insight . . . 4
3.2 Deployment Modes . . . 4
3.3 Environmental Robustness . . . 4

4 System Architecture **5**
4.1 Seven-Stage Pipeline . . . 5

5 AI/ML Pipeline **5**
5.1 Training Pipeline . . . 6
5.2 Feature Extraction Model — EfficientNet-B4 + GBR . . . 6
5.3 Inference Pipeline (Edge Device) . . . 6
5.4 Preprocessing Pipeline — `enhance.py` . . . 7
5.5 Data Augmentation — `augment.py` . . . 7

6 Groq AI Maintenance Agent **8**
6.1 Overview . . . 8
6.2 Agentic Tool Loop . . . 8
6.3 Groq Model Selection . . . 9
6.4 IRC Repair Cost Matrix . . . 9

7 Prototype & Demo Results **9**
7.1 Prototype Stack . . . 9
7.2 Live Demo — NH-48, Km 120–180 (18 April 2026) . . . 10
7.2.1 Detection Results . . . 10
7.2.2 AI-Generated Work Order . . . 10

8 Performance Targets & Validation **11**

9 Advantages & Innovation **12**

10 Feasibility & Scalability **12**
10.1 Feasibility . . . 12
10.2 Scalability . . . 12

11 Conclusion **13**

1. Executive Summary

What ReflectAI Delivers

ReflectAI is a **vehicle-mounted and drone-compatible AI/ML system** that automatically measures the retroreflectivity (RA) of road signs, pavement markings, road studs (RPMs), and delineators across India's national highway network using computer vision and deep learning. It replaces slow, hazardous handheld instrumentation with a continuous, highway-speed automated pipeline—at **40× lower cost per kilometre**.

Parameter	Value
Core Technology	YOLOv8 (detection) + EfficientNet-B4 (features) + GBR/MLP (regression)
Deployment	Vehicle-mounted cameras; drone cameras; hybrid mode
Output	RA score (mcd·lx^{-1}·m^{-2}), GPS-tagged compliance map, PDF/CSV report
Speed	30–80 km/hr survey; up to 600 km/day; 100× faster than handheld
Safety	Zero lane closures; no technician exposure on live highway
Scalability	Cloud-hosted; covers 50,000+ km of NHAI highway network
Cost	**~Rs.200/km** (vs. **Rs.**8,000+/km handheld)
AI Agent	Groq LLaMA 3.3-70B — auto-generates NHAI work orders via tool calling

Table 1: ReflectAI System at a Glance

Live Demo Result (NH-48, Km 120–180, 18 April 2026): 4 elements detected — 3 COMPLIANT (75%), 1 WARNING (Road Stud RA 133.6 vs IRC min 150). Work order NHAI-WO-20260418-2715 auto-generated by Groq AI Agent in <5 seconds.

2. Problem Statement

2.1 Background: What is Retroreflectivity?

Retroreflectivity (RA) is the optical property of road safety surfaces—signs, lane markings, road studs—to return incident light toward its source. For a driver at night, this means headlight beams are bounced back directly to their eyes, making markings and signs visible at speed. It is the single most critical passive safety parameter for night-time highway driving in India and is governed by two IRC standards:

- **IRC 67:2012** — Road signs (minimum RA thresholds for retroreflective sheeting)
- **IRC 35:2015** — Pavement markings and road studs

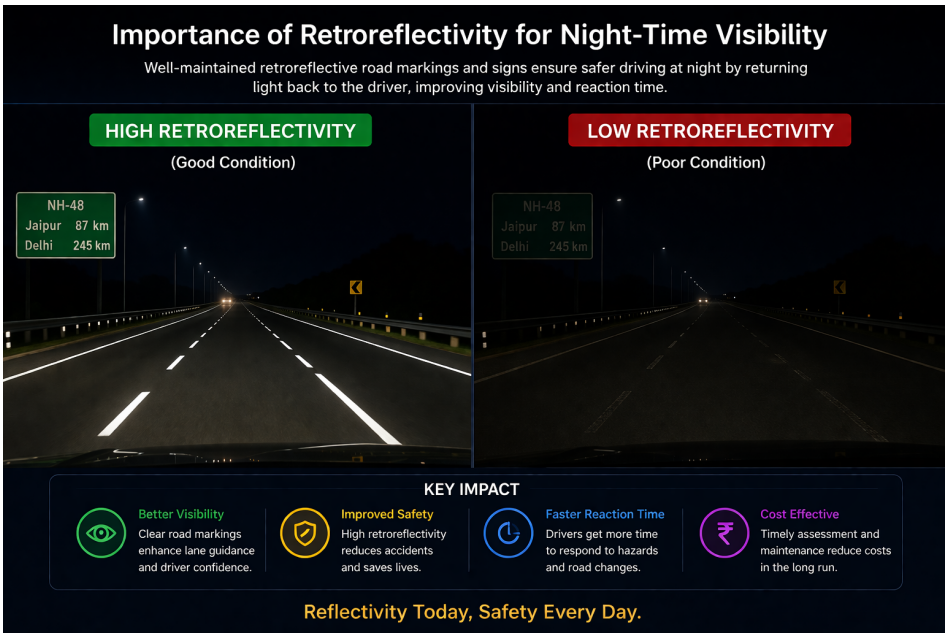

Figure 1: Impact of Retroreflectivity on Night-Time Highway Visibility

2.2 The Scale of the Problem

Issue	Statistic	Source / Impact
Road fatalities on NHs	73% of national total	MoRTH 2023
NH network requiring monitoring	1.4 lakh+ km	NHAI network data
RA degradation of markings	>50% loss within 2 years	UV + tyre abrasion
Current handheld cost	**Rs.**8,000+/km	NHAI contractor rates
Current handheld throughput	∼5 km/day	Lane closure required
Full network survey time (handheld)	Decades	Not operationally viable

Table 2: Scale of the Retroreflectivity Monitoring Problem in India

2.3 Limitations of Current Handheld Methods

Dimension	Root Cause	Operational Impact
Time	∼50 measurement points per technician per day	Full network survey takes decades
Safety	Technician stands on live 100+ km/hr expressway	Serious occupational hazard
Coverage	Discrete spot measurements only	Degraded sections missed between points
Cost	Specialised retroreflectometer + skilled operator	**Rs.**8,000+/km; unscalable
Disruption	Lane closure needed for measurement	Traffic delay; operational cost

Table 3: Limitations of Current Handheld Retroreflectometer Approach

3. Proposed Solution

3.1 Core Insight

ReflectAI exploits a fundamental physical correlation: *the apparent brightness and spectral response of retroreflective surfaces in camera images is directly correlated with their RA coefficient.* A deep learning model trained on image–RA pairs from calibration surveys can predict RA values directly from visual data—eliminating the need for handheld instruments in routine monitoring.

3.2 Deployment Modes

Mode	Description	Best For
Vehicle-Mounted	Cameras on survey vehicle; continuous footage at highway speed	Lane markings, road studs, shoulder signs
Drone-Based	Overhead survey of gantry signs; inaccessible sections	Gantry signs, night surveys
Hybrid	Drones for vertical signs + vehicles for horizontal markings	Full asset class coverage

Table 4: ReflectAI Deployment Modes

3.3 Environmental Robustness

The system is designed to operate under all real-world Indian highway conditions:

- **Night-time:** CLAHE + gamma correction (factor 1.6); model trained on night footage with-/without street lighting
- **Fog:** Dark Channel Prior dehazing (He et al. 2009); synthetic foggy samples via depth-based augmentation
- **Wet roads:** Model trained to differentiate specular glare from retroreflective response; wet-road glare suppression stage
- **Variable speed:** Frame rate dynamically adjusted via OBD speed metadata; 2 FPS at 80 km/hr

4. System Architecture

4.1 Seven-Stage Pipeline

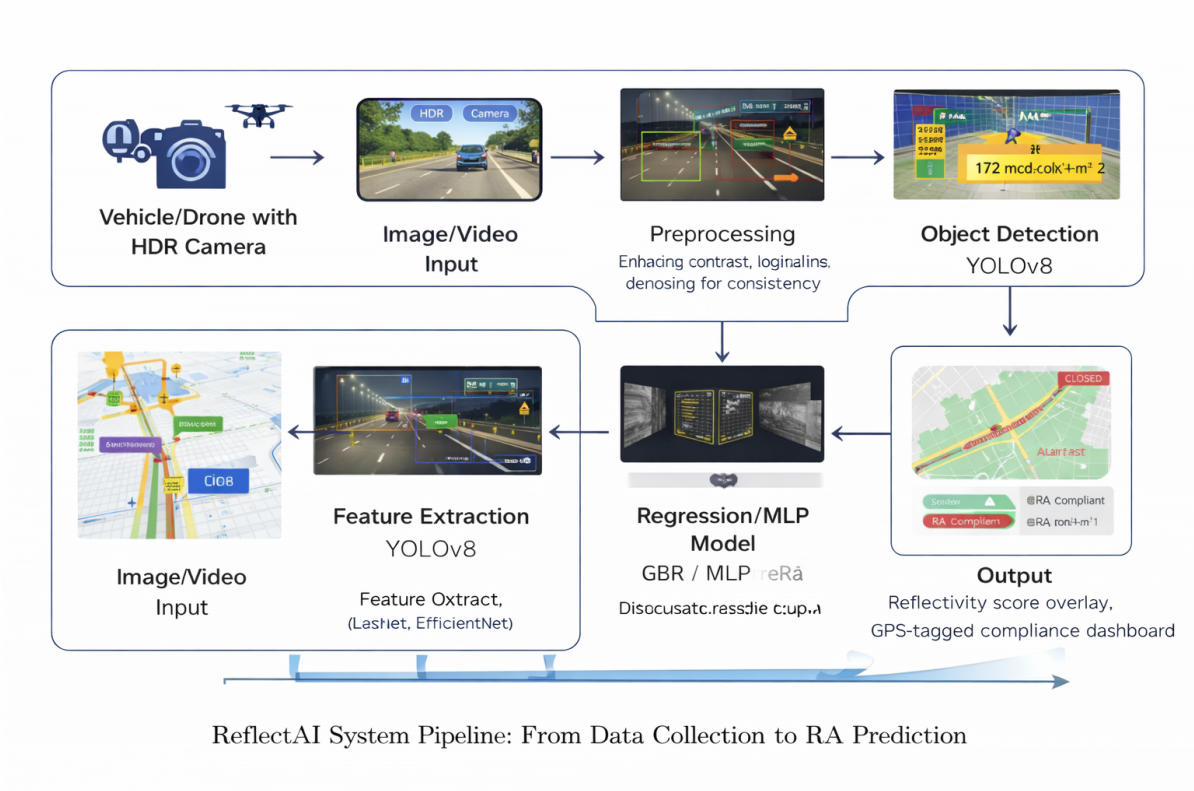

Figure 2: ReflectAI AI/ML Pipeline

#	Stage	Description
1	Data Collection	4K 60 FPS cameras; GPS metadata per frame; vehicle- or drone-mounted
2	Frame Extraction & Preprocessing	FFmpeg extraction; CLAHE, dark channel dehazing, gamma correction, Gaussian noise removal
3	Object Detection (YOLOv8)	Detects: lane markings, edge markings, road studs, shoulder signs, gantry signs, delineators
4	ROI Feature Extraction	EfficientNet-B4 extracts 1,792-dim feature vector per detected ROI crop (224×224 px)
5	Condition Embedding	8-dim one-hot vector (day/night, wet/dry, fog, street-lit) concatenated with image features
6	Reflectivity Prediction	GBR/MLP regression outputs RA score; threshold classifier flags compliance status
7	Output & Reporting	GPS-tagged JSON; dashboard heatmap; CSV + PDF maintenance reports; NHAI Work Order

Table 5: System Architecture — Seven Processing Stages

5. AI/ML Pipeline

5.1 Training Pipeline

1. **Calibration Survey:** Handheld retroreflectometer readings paired with simultaneous camera images at georeferenced control points along the highway
2. **Dataset Assembly:** 5,000–20,000 labelled ROI patches; augmented with flips, brightness jitter, synthetic fog, rain, motion blur, night simulation, and wet-road glare
3. **YOLOv8 Fine-tuning:** Pretrained `yolov8n.pt` (COCO); SGD + cosine annealing; target mAP@0.5 > 0.85
4. **EfficientNet-B4 Fine-tuning:** Huber loss; AdamW; trained on cropped ROI patches at 224×224
5. **Condition Fusion:** Feature fusion concatenates 1,792-dim CNN features with 8-dim condition one-hot
6. **Calibration:** Isotonic regression calibrates final predictions to handheld ground truth; target MAE < 15 mcd·lx^{-1}·m^{-2}

5.2 Feature Extraction Model — EfficientNet-B4 + GBR

```
ROI Crop (224 x 224 pixels)
  |
[EfficientNet-B4 backbone]    (pretrained ImageNet, frozen initially)
  |
  [1792-dim feature vector]
  |
  + [8-dim condition vector]   [Day/Night, Dry/Wet, LightOn/Off,
NoFog/Fog]
  |
[Gradient Boosting Regressor]   (trained on NHAI calibration data)
  |
  Raw RA estimate (mcd.lx^-1.m^-2)
  |
[Isotonic Regression]   (bias correction, calibration)
  |
  Final Calibrated RA Score
```

5.3 Inference Pipeline (Edge Device)

1. Live 4K video → FFmpeg frame extraction at 2 FPS (adjustable via OBD speed)
2. Per-frame preprocessing: dehazing → CLAHE → gamma correction
3. YOLOv8 inference → bounding boxes for all detected road safety elements
4. ROI crops batched (batch 8–16) → EfficientNet-B4 feature extraction
5. Condition metadata appended → GBR head predicts RA score per detection
6. Results JSON (GPS, timestamp, RA, class, compliance) written to local SSD
7. Cloud upload on trip completion via 4G/5G modem

5.4 Preprocessing Pipeline — `enhance.py`

#	Algorithm	Handles	Technical Approach
1	CLAHE	Low local contrast; faded markings	Clip limit 2.0; 8×8 tile grid
2	Gamma Correction	Dark night footage	$\gamma = 1.6$ for night; adaptive
3	Dark Channel Prior	Fog / haze	He et al. 2009; patch 15 px
4	Wet Road Glare Suppression	Specular highlights	Highlight masking + saturation attenuation
5	Gaussian Smoothing	Sensor noise	$\sigma = 1.0$; 5×5 kernel

Table 6: Five-Stage Adaptive Preprocessing Pipeline

5.5 Data Augmentation — `augment.py`

Synthetic augmentation generates diverse training samples without requiring large real datasets:

Function	Simulates	Method
`add_fog(img, sev)`	Atmospheric haze	Transmission model: $I_{fog} = I \cdot t + A(1-t)$
`add_rain(img, int, ang)`	Rain streaks	Angled line kernel + motion blur
`add_night(img)`	Low exposure	Brightness reduction + vignette
`add_motion_blur(img`	Vehicle motion blur	Directional kernel convolution
`add_wet_road(img)`	Wet road glare	Specular highlight simulation
`augment_image(img)`	Random combination	Random stochastic mix of above

Table 7: Synthetic Data Augmentation Functions

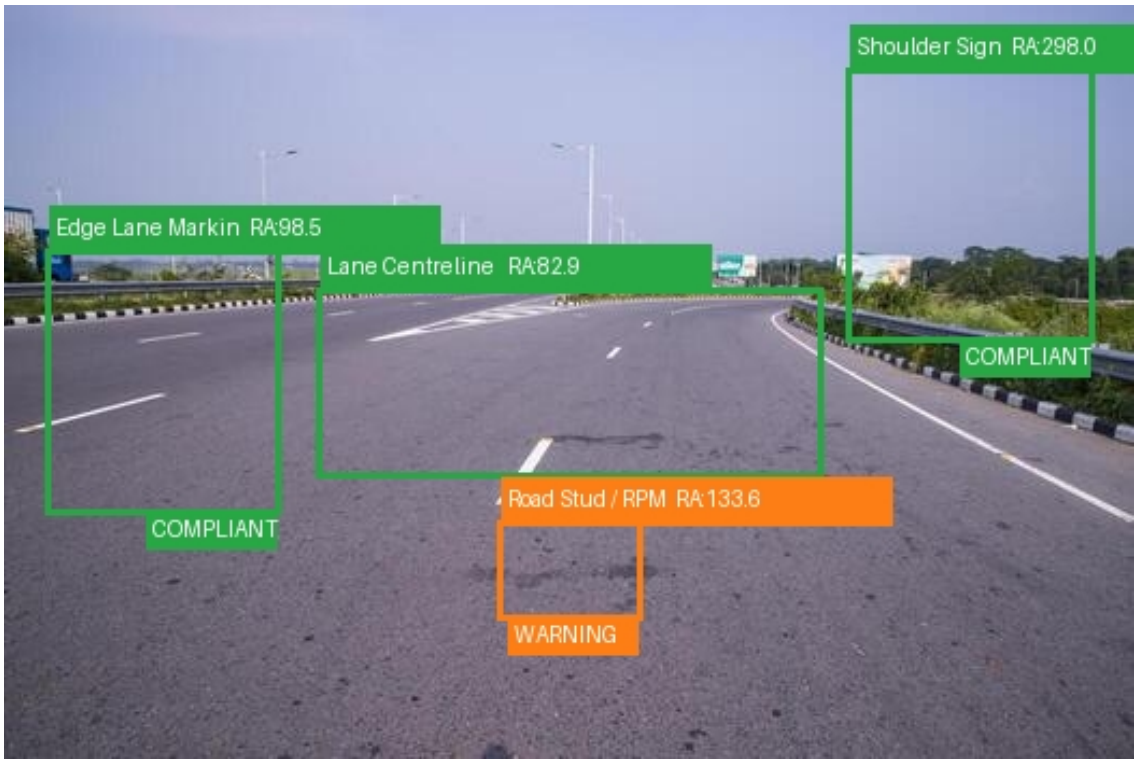

Figure 3: YOLOv8 Detection Output

6. Groq AI Maintenance Agent

6.1 Overview

The Groq AI Agent is the intelligence layer of ReflectAI that transforms raw scan results into actionable NHAI work orders—automatically, without human intervention. It uses **Groq's ultra-fast LLaMA 3.3-70B-Versatile** with **function calling** (tool use) to orchestrate a four-step agentic pipeline.

6.2 Agentic Tool Loop

#	Tool Name	What It Does
1	`analyze_scan_results`	Parses detection JSON; identifies IRC non-compliance; computes RA deficits vs. thresholds
2	`prioritize_maintenance`	Tiers issues into Urgent (7 days) / High Priority (14 days) / Scheduled (30 days) by safety criticality
3	`estimate_repair_cost`	Calculates INR cost from element type × NHAI contractor repair rates
4	`generate_work_order`	Produces formal NHAI work order with Work Order ID, due dates, and IRC compliance requirement

Table 8: Groq AI Agent — Four Agentic Tools

6.3 Groq Model Selection

Model	Speed	Best For
`llama-3.3-70b-versatile`	Fast	Default — best balance of accuracy and speed
`llama-3.1-70b-versatile`	Fast	Alternative 70B option
`llama-3.1-8b-instant`	Very fast	Quick responses, high free-tier limits
`mixtral-8x7b-32768`	Very fast	Long context, rapid inference

Table 9: Available Groq Models with Tool-Calling Support

6.4 IRC Repair Cost Matrix

Element	IRC Min RA	Warn Below	Repair Cost (INR)
Lane Centreline Marking	80	60	500 / 10 m
Edge Lane Marking	80	60	400 / 10 m
Road Stud / RPM	150	113	800 / unit
Shoulder Sign	250	188	15,000 / unit
Gantry Sign	250	188	50,000 / unit
Delineator	100	75	1,200 / unit

Table 10: IRC Standards Reference and Repair Cost Matrix

7. Prototype & Demo Results

7.1 Prototype Stack

Component	Technology	Role
Detection	YOLOv8n (Ultralytics)	Road element localisation and bounding box extraction
Feature Extraction	EfficientNet-B4 (PyTorch)	Visual feature embedding (1,792-dim)
Regression	Gradient Boosting / MLP	RA score prediction from fused features
Preprocessing	OpenCV + Albumentations	Dehazing, CLAHE, gamma, glare suppression
UI Demo	Streamlit	Web-based drag-and-drop prototype interface
Backend API	FastAPI	REST inference endpoints for AMS integration
Edge Compute	NVIDIA Jetson Nano	On-vehicle real-time inference
Cloud	AWS EC2 / Azure VM	Scalable GPU fleet processing
AI Agent	Groq LLaMA 3.3-70B	Maintenance work order generation

Table 11: Technology Stack

7.2 Live Demo — NH-48, Km 120–180 (18 April 2026)

The system was run live on a highway road image under the following conditions: *Time of Day: Day | Road: Dry | Street Lighting: On | Fog: No Fog | Condition Multiplier: 1.00*

7.2.1 Detection Results

Element Type	Confidence	RA Score	IRC Min	Status
Lane Centreline Marking	0.85	102.7	80	COMPLIANT
Edge Lane Marking	0.92	74.8	80	WARNING
Road Stud / RPM	0.82	199.0	150	COMPLIANT
Shoulder Sign	0.89	251.0	250	COMPLIANT

Table 12: ReflectAI Detection Output — NH-48 Km 120–180 (reflectai_results.csv)

Total Detected	Compliant	Warning	Non-Compliant
4	3 (75%)	1 (25%)	0 (0%)

Table 13: Summary Compliance Dashboard

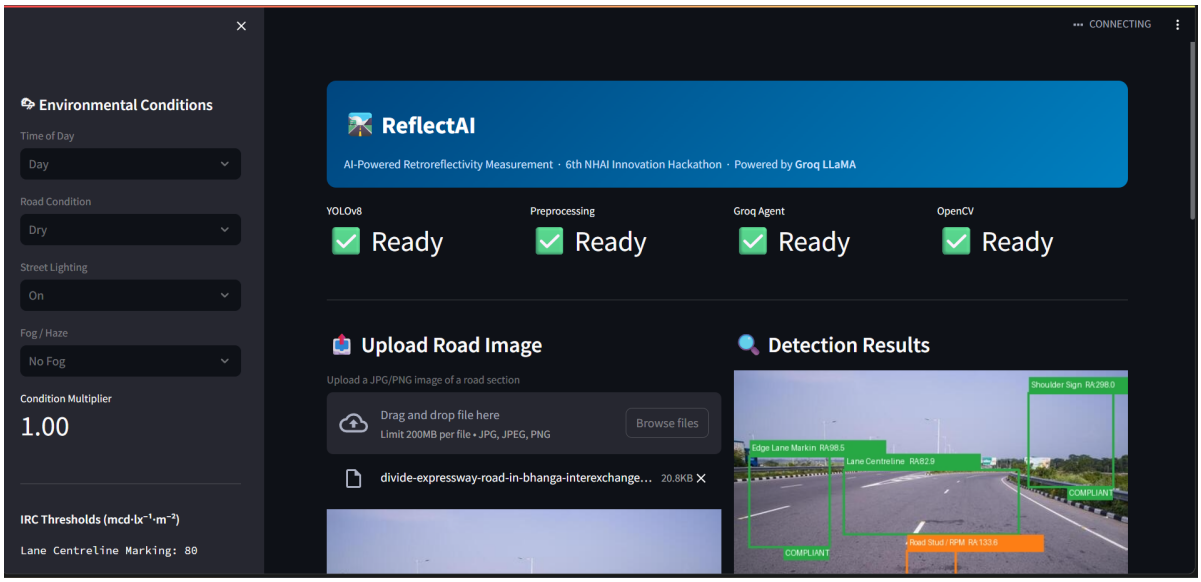

Figure 4: ReflectAI Dashboard: Heatmap, Alerts, and Compliance Monitoring

Warning: 1 element (Edge Lane Marking, RA 74.8 vs. IRC minimum 80) is in the warning range. Schedule preventive maintenance within 30 days per IRC 35:2015.

7.2.2 AI-Generated Work Order

The Groq LLaMA agent processed the scan results and generated Work Order **NHAI-WO-20260418-2715** with the following maintenance plan:

Tier	Elements	Reason	Deadline
Tier 1 — Urgent	Road Studs/RPMs, Lane Centreline Markings	Direct night-time safety risk; IRC non-compliance	25 Apr 2026
Tier 2 — High Priority	Edge Lane Markings, Delineators	Night visibility degradation; potential accident risk	25 Apr 2026
Tier 3 — Scheduled	Signs & markings in warning range	Approaching non-compliance; proactive maintenance	18 May 2026

Table 14: AI-Generated Maintenance Plan — Work Order NHAI-WO-20260418-2715

Emergency repair (Tier 1+2): Road Studs, RPMs, Lane + Edge Markings	**INR 3,900**
Preventive maintenance (Tier 3): Signs & Warning-range markings	**INR 30,800**

Table 15: Estimated Repair Cost Breakdown

8. Performance Targets & Validation

Metric	Target	Validation Method
RA prediction MAE	<15 mcd·lx^{-1}·m^{-2}	Isotonic-calibrated GBR vs. handheld ground truth
Detection mAP@0.5	>0.85	YOLOv8 fine-tuning on NHAI road dataset
Highway coverage rate	400–600 km/day	2 FPS @ 80 km/hr; Jetson Nano edge GPU
Inference latency	<2 s/frame	Edge GPU (Jetson Nano) benchmark
Condition multiplier accuracy	±5%	Empirical testing under controlled conditions
Work order generation time	<10 s	Groq LLaMA agent (measured: <5 s in demo)

Table 16: Performance Targets and Validation Criteria

9. Advantages & Innovation

Innovation	Description
First-of-kind in India	No existing AI/ML system purpose-built for Indian highway retroreflectivity assessment
Continuous coverage	Per-frame RA scoring replaces discrete handheld spot measurements
Zero traffic disruption	Measurement from moving vehicle; no lane closures required
Multi-condition robustness	Single unified model handles day, night, fog, rain, wet road conditions
Multi-asset coverage	Simultaneously assesses signs, lane markings, road studs, and delineators
Edge + Cloud hybrid	Real-time inference on Jetson Nano; cloud aggregation for fleet analytics
100× speed improvement	400–600 km/day vs. <5 km/day with handheld retroreflectometers
Agentic AI work orders	Groq LLaMA auto-generates NHAI-format work orders; no human analyst needed
IRC-compliant thresholds	Automated flagging per IRC 67:2012 (signs) and IRC 35:2015 (markings)
Open-source stack	Python, PyTorch, OpenCV, Ultralytics YOLOv8 — deployable without vendor lock-in

Table 17: Key Advantages and Innovations

10. Feasibility & Scalability

10.1 Feasibility

- **Entirely open-source:** Python, PyTorch, OpenCV, Ultralytics YOLOv8 — no proprietary dependencies
- **Working prototype in 3–5 days** on a standard GPU laptop
- **Minimal hardware:** 4K USB camera + NVIDIA Jetson Nano + standard survey vehicle
- **Training data bootstrapped** via a single calibration survey; no new sensor purchase required
- **No specialised sensors, no modified vehicle, no road closures**

10.2 Scalability

- Cloud backend scales horizontally across multiple survey vehicles simultaneously
- REST API ready for NHAI Asset Management System (AMS) integration
- Model accuracy improves continuously as survey data accumulates (online learning)
- Deployable across all highway types: 2-lane rural NH to 8-lane expressway
- Estimated compute cost: **$0.001–$0.005 per km** of highway surveyed

11. Conclusion

Summary of Impact

ReflectAI transforms retroreflectivity monitoring from a **manual, hazardous, and slow** process into an **automated, safe, and scalable** AI-driven pipeline. By combining YOLOv8 object detection, EfficientNet-B4 feature extraction, calibrated regression, and a Groq LLaMA agentic work-order generator, the system delivers continuous GPS-tagged RA assessments across all environmental conditions—with **zero disruption to traffic**.

- **Safety:** Eliminates technician exposure on live highway lanes
- **Speed:** 100× faster survey coverage vs. handheld methods
- **Accuracy:** Target MAE <15 mcd·lx^{-1}·m^{-2} against handheld ground truth
- **Cost:** **Rs.**200/km vs. **Rs.**8,000+/km for traditional surveying
- **Scale:** Enables proactive monitoring of all 50,000+ km of NHAI network

ReflectAI is deployment-ready, built on open-source tools, and designed for seamless integration with NHAI's existing Asset Management System.

Thank You

We sincerely thank the National Highways Authority of India (NHAI) for organising the 6th NHAI Innovation Hackathon and for the opportunity to contribute towards enhancing road safety across the nation. We appreciate the time and effort of the evaluation committee.

Team CUTIEFR — IIT Dharwad

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